Supplementary material for "Strain-coupled one-dimensional turbulence: a single-line closure for rapid distortion, its measured limits, and the consequence for leading-edge noise"

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This supplement carries two blocks of supporting material referenced from the main text: the validation against the strained-turbulence experiment of Chen *et al.* (2006) (§ S1), summarised in the main text's validation section, and the complete intervention family behind the scale-local-relaxation ledger of the main text's measured-limits section (§ S2).

S1. Validation against the strained-turbulence experiment of Chen, Meneveau & Katz

The strain-on/strain-off comparison of the main text is model-internal; an external benchmark must confirm that both the rapid-distortion limit and the departure from it are physical. We use the plane-strain experiment of Chen *et al.* (2006) (CMK), in which initially near-isotropic turbulence at $R_\lambda \approx 400$ is subjected to a controlled planar straining cycle and the one-dimensional component spectra are compared with rapid-distortion theory. We reproduce the *straining* phase of their cycle (the destraining phases involve a strain reversal outside the present monotonic configuration); its end corresponds to a total deformation $D = \exp(\frac{1}{2}\int S\,\mathrm{d}t) \approx 3$, i.e. $e \approx 2.20$ in the notation of the main text.

In the rapid-distortion limit (eddy events suppressed, inviscid) the model must reproduce the analytic reference of Lee *et al.* (1986) that CMK use. Figure S1 shows that it does: the component spectral centroids follow the rigid-translation law, reaching the deformation ratio 3.00 at $e = 2.20$ with the components coincident, and the kinetic energy reaches 1.72 against the exact rapid-distortion amplification 1.69—the residual being the moment closure's, not the implementation's (main text, verification section), since with the events suppressed the solver evolves the closure trajectory (1.72) exactly. The truly inviscid setting is required: any nonzero viscosity dissipates the compressed small scales and degrades the rigid translation, an effect we verified directly.

With the eddy events and viscosity active the model is no longer constrained to rigid translation, and the strain drives a measurable anisotropy between the components. Figure S2 shows the strain-induced migration of the streamwise and upwash component centroids, each measured against the matched no-strain baseline (the per-component form of the strain-on/strain-off construction, which cancels the initialisation transient and the eddies' own relaxation): the components diverge progressively under continued strain—

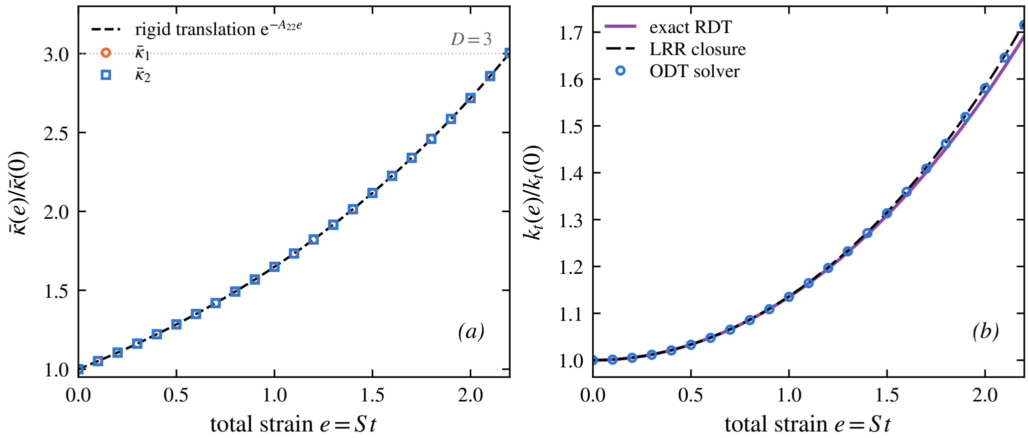

Figure S1. Rapid-distortion limit (eddy events suppressed, inviscid) against the analytic reference of Lee *et al.* (1986) used by Chen *et al.* (2006). (*a*) Component spectral-centroid migration follows the rigid-translation law, reaching the deformation ratio $D = 3$ at the end of the straining phase. (*b*) Kinetic-energy amplification: the solver follows the closure trajectory, within 2% of the exact rapid-distortion value at $e = 2.20$.

the streamwise migration reaching 1.9 at the end of the straining phase against 1.5 for the upwash, both far below the rigid value 3—so the upwash component is held back most. This is the qualitative behaviour CMK report—the strained spectrum departs from the isotropic-translation expectation with the components responding differently—and it is the spectral signature of the broadband distortion analysed in the main text. The comparison is restricted to its qualitative content, for three reasons stated plainly: the band-limited initial spectrum differs from CMK's, so absolute levels are not comparable; the line-resolved one-dimensional spectra and the experiment's transverse/longitudinal projections are not the same projection of the spectrum tensor; and the scale separation achieved here ($\mathcal{L}/\eta \approx 250$) is below the experimental ≈ 930, whose compressed dissipation range exceeds feasible line resolution. Within these limits the model reproduces the rapid-distortion reference and the qualitative departure mechanism of the benchmark; the quantitative spectral assessment is carried out in the main text against the strained-box DNS, where the projection and the initial spectrum are matched by construction. A direct numerical simulation of the same straining–destraining cycle (Gualtieri & Meneveau 2010), at a Reynolds number below the experiment's, reports the component variances and the pressure–strain budget through the cycle and compares them with a Launder–Reece–Rodi closure; it is the DNS counterpart against which the present model can be set quantitatively on this configuration. A direct overlay is not attempted here: their observable is the anisotropy through the full straining–*destraining* cycle, whereas the monotonic straining phase reproduced above does not include the strain reversal, so the comparison is kept at the level of the independent confirmation their DNS provides for the experiment.

S2. The complete intervention family

Table S1 records every intervention of the measured-limits campaign—the knob, its measured fine-scale effect on the transmission diagnostic, and the reading—and figure S3 shows the eddy-locked family on all four diagnostic panels. Each case was run at 1024 realisations with seeds paired to a common baseline; effects are median paired same-seed contrasts, with "significant" meaning the 95% bootstrap interval excludes zero. The

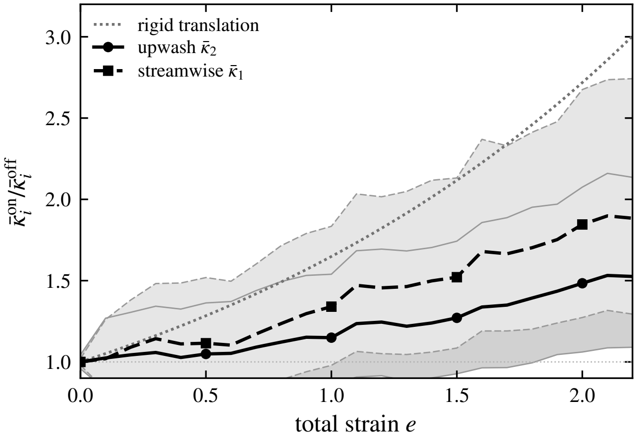

Figure S2. Full model (eddies and viscosity active; ensembles over 128 realisations at $\nu = 10^{-5}$): strain-induced migration of the streamwise and upwash component centroids, each as the ratio of the strained ensemble to the matched no-strain baseline, against the rigid-translation law (grey dotted). The components diverge under strain—the upwash held back most—reproducing the qualitative departure-from-RDT mechanism of Chen *et al.* (2006). The comparison is qualitative; see text.

main text carries the ledger these results establish and the two figures that decide it (the relaxation-clock figure and the five-way benchmark comparison).

S3. The relaxation clock in the strained spectra

Figure S4 shows the spectral counterpart of the transmission-diagnostic results of the main text, on the Gate-A production ensembles (1024 realisations per case and variant, both rapidities): the size-capped relaxation clock roughly halves the distance between the model's strained line spectra and the exact rapid-distortion reference family, and—unlike the standard model—holds that distance *constant in strain*. The frozen isotropic fit included for reference is the spectral content of any frozen-inflow practice: unable to represent the strained anisotropy at all, its distance grows to 45–51% by $e = 2$. The component-ratio panel localises the improvement: exact rapid distortion drives the fine scales towards component equality, the standard model over-transmits the anisotropy there, and the clock moves the measured ratio towards the linear reference precisely below its size cap.

References

Chen, Jun, Meneveau, Charles & Katz, Joseph 2006 Scale interactions of turbulence subjected to a straining–relaxation–destraining cycle. *Journal of Fluid Mechanics* **562**, 123–150.

Gualtieri, P. & Meneveau, C. 2010 Direct numerical simulations of turbulence subjected to a straining and destraining cycle. *Physics of Fluids* **22** (6), 065104.

Lee, Moon J., Piomelli, Ugo & Reynolds, William C. 1986 Useful formulas in the rapid distortion theory of homogeneous turbulence. *Physics of Fluids* **29** (10), 3471–3474.

intervention (knob)	fine-scale effect	reading
acceptance gate on post-event anisotropy (45–98% rejection)	null at all thresholds; harshest drifts to eddy-free limit	gate is scale-blind: modulates eddy rate, not the scale allocation
gate on sub-band eddies only (42%, 97% rejection)	null, including band *energy*	fine scales are fed by large-eddy maps directly
unequal map images (middle $\frac{2}{3}$)	null (one-point statistics move: $\mathcal{A}_{lo}$ 2.85 $\rightarrow$ 2.50 at $e = 3.9$)	outer images feed the band in one step; channels trade off
sub-scale kernels, 1 level ($\ell/3$)	null within CIs	$\ell/3$ sits above the measured band
sub-scale kernels, 2 levels ($\ell/9$)	$\mathcal{A}_{hi}$: 1.43 $\rightarrow$ 1.27 at $e = 1$ (significant, both bands)	first significant fine-scale isotropisation
sub-scale kernels, 3 levels ($\ell/27$)	significant at $e = 1$ *and* $e = 3$ (-12% paired)	deepest levels reach the band
depth tied to eddy size	significant in both bands at $e = 1$; no late-strain reversal	best of the family: gains without the deep-strain overshoot
2 levels iterated 3× per event	strongest at $e = 1$ (1.29); overshoots at deep strain	per-event amplitude is not the limiter
concurrent relaxation: N same-size intervals, randomly placed ($N = 1$, 3)	$N = 1$ null; $N = 3$ significant in both bands at $e = 1$, strongest at the energy-containing scales; fade persists	the sampled intervals act at eddy scale; delocalising the events alone is not the lever
concurrent + depth ($N = 3$, adaptive hierarchy in each interval)	strongest early effect of the eddy-locked family ($\mathcal{A}_{hi}$ 1.43 $\rightarrow$ 1.26); low band significant even at $e = 3.9$; fine scales overshoot depth-only at deep strain	dosage helps early and at large scales, and harms deep-strain fine scales wherever it is applied
independent relaxation clock, uncapped (rate matched to the eddy rate)	significant in both bands at *every* strain, growing with strain—the fade broken—but the one-point anisotropy is suppressed ($u_2^2/2k_t$ 0.58 $\rightarrow$ 0.41 at $e = 3.9$)	event-locked timing was the deep-strain limiter; a scale-blind stream fights the strain itself
independent clock, size-capped ($\ell \leq \ell^*$, adaptive depth inside)	fine scales significant at every strain with no fade ($\mathcal{A}_{hi}$ held at 1.30–1.44); one-point trajectory intact for $e \lesssim 2$, mildly reduced beyond	timing + scale selectivity: the working form of the mechanism

Table S1. The intervention family on the transmission diagnostic (1024 paired-seed realisations per case; effects quoted as median paired same-seed contrasts, "significant" meaning the 95% bootstrap interval excludes zero).

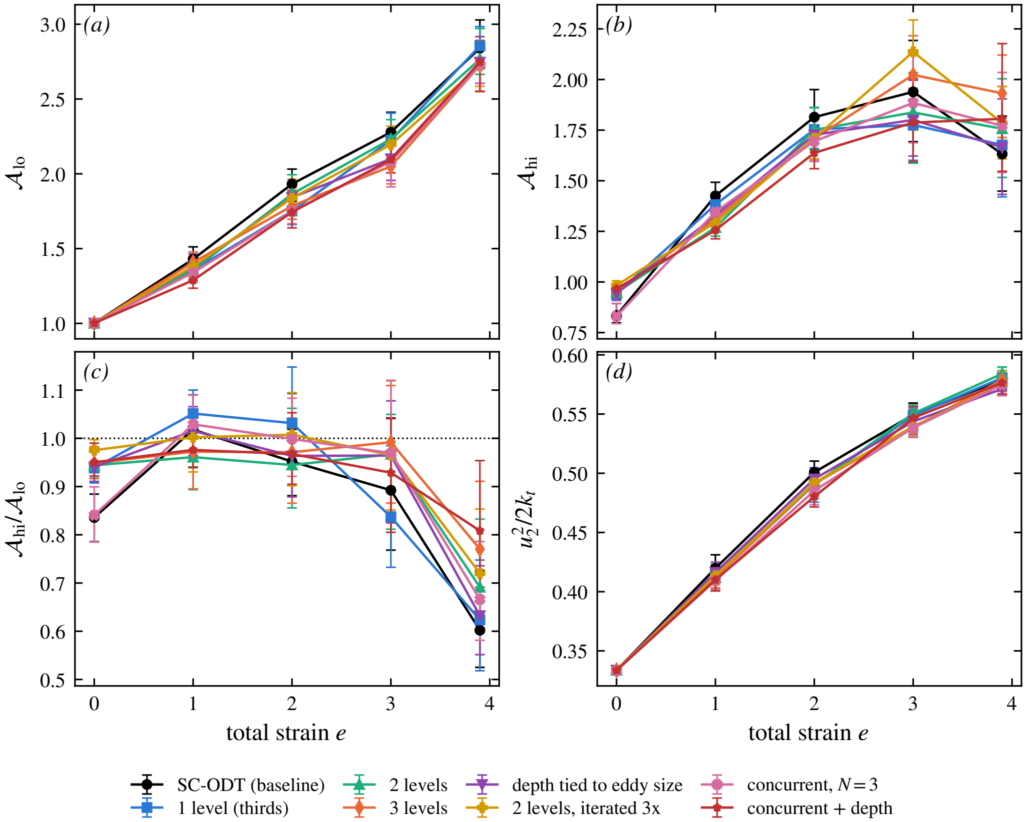

Figure S3. Scale-local relaxation on the transmission diagnostic: (*a*) energy-containing-band imbalance $\mathcal{A}_{\mathrm{lo}}$, (*b*) fine-scale-band imbalance $\mathcal{A}_{\mathrm{hi}}$, (*c*) transmission $\mathcal{A}_{\mathrm{hi}}/\mathcal{A}_{\mathrm{lo}}$, (*d*) upwash energy fraction, against total strain: standard ODT, hierarchical sub-scale kernels at fixed depth (1–3 levels), depth tied to eddy size, the per-event iterated variant, concurrent relaxation (N = 3 randomly placed same-size intervals per event), and the stacked variant (concurrent intervals each running the adaptive hierarchy). Medians over 1024 paired-seed realisations, 95% bootstrap intervals.

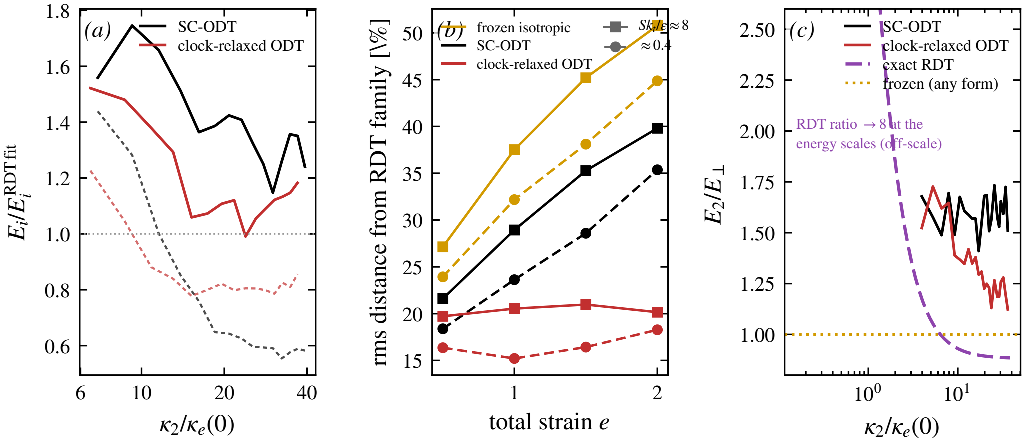

Figure S4. (a) Binned component spectra over their best exact-RDT-family curves at $e = 2$, $Sk_t/\varepsilon \approx 8$ (solid: E_2; dashed: $E_\perp$): the standard model's misallocation (upwash high, transverse low) and its reduction by the clock. (b) rms log-distance from the family against total strain at both rapidities: frozen isotropic representation (gold), standard model (black), size-capped clock (red) — the clock's distance is strain-independent. (c) Component ratio $E_2/E_\perp(\kappa_2)$ at $e = 1$, rapid: measured standard and clock ensembles against the exact-RDT reference; any frozen form predicts the ratio 1 identically.